

1-13 The 12-V automobile battery in Figure P1-13 has an output capacity of 100 ampere-hours (Ah) when connected to a head lamp that absorbs 200 watts of power. The car engine is not running and therefore not charging the battery. Assume the battery voltage remains constant.

- Find the current supplied by the battery and determine how long can the battery power the headlight.
- A 100 W device is connected through the utility port. How long can the battery power both the headlight and the device?

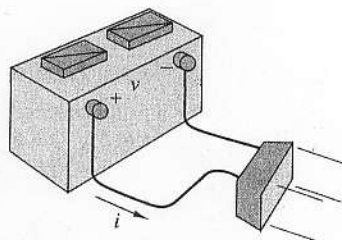


FIGURE P1-13

1-14 An incandescent lamp absorbs 100 W when connected to a 120-V source. A energy-efficient compact fluorescent lamp (CFL) producing the same amount of light absorbs 16 W when connected to the same source.

How much cheaper is it to operate the CFL versus the incandescent bulb over 1,000 hours when electricity costs 7.8 cents/kWh?

1-21 Two electrical devices are connected as shown in Figure P1-21. Using the reference marks shown in the figure, find the power transferred and state whether the power is transferred from A to B or B to A when

- $v = +11$ V and $i = -1.1$ A
- $v = +80$ V and $i = +20$ mA
- $v = -120$ V and $i = -12$ mA
- $v = -1.5$ V and $i = -600$ μ A

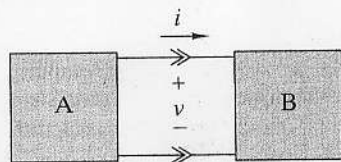


FIGURE P1-21

2-3 A 100-k Ω resistor dissipates 100 mW. Find the current through the resistor.

2-5 In Figure P2-5 the resistor dissipates 25 mW. Find R_x .

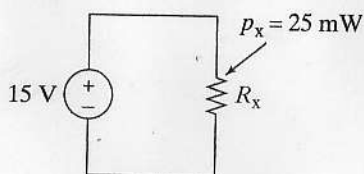


FIGURE P2-5

2-12 A thermistor is a temperature-sensing element composed of a semiconductor material that exhibits a large change in resistance proportional to a small change in temperature. A particular thermistor has a resistance of 5 k Ω at 25 $^{\circ}$ C. Its resistance is 340 Ω at 100 $^{\circ}$ C. Assuming a straight-line relationship between these two values, at what temperature will the thermistor's resistance equal 1 k Ω ?

2-13 In Figure P2-13 $i_2 = -5$ A and $i_3 = 2$ A. Find i_1 and i_4 .

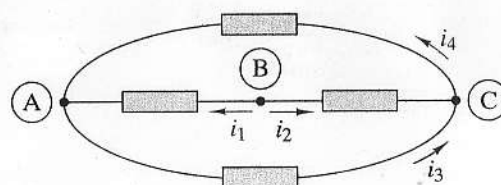


FIGURE P2-13

2-14 In Figure P2-14 $v_1 = 3$ V and $v_3 = 5$ V. Find v_2 , v_4 and v_5 .

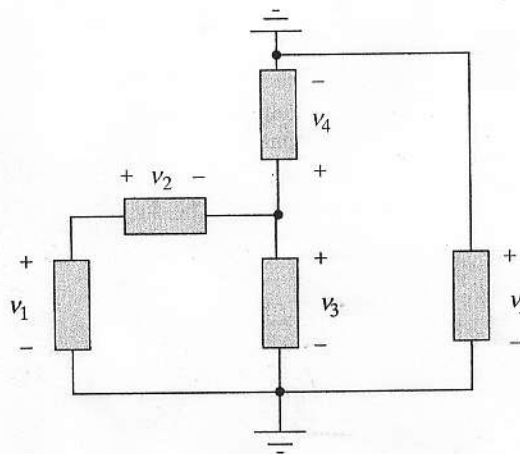


FIGURE P2-14

2-15 For the circuit in Figure P2-15:

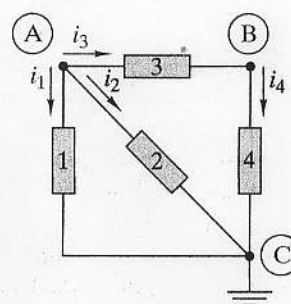


FIGURE P2-15

2-21 In Figure P2-21 $v_2 = 10$ V, $v_4 = 5$ V, and $v_5 = 15$ V. Find v_1 , v_3 , and v_6 .

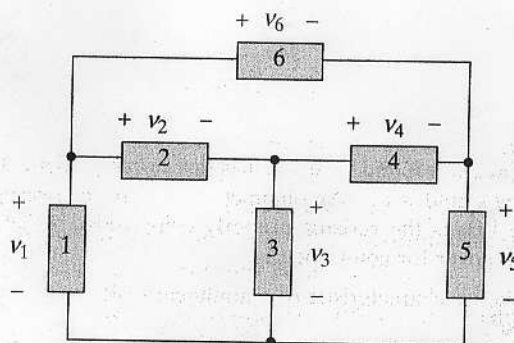


FIGURE P2-21